

BTech LEET - System Architecture Document (v1)

Project Goal

Build the largest platform for Indian Diploma-to-BTech lateral entry (LEET) aspirants.

The platform should allow students to:

- Discover LEET exams
- Track notifications
- Download previous year papers
- Read preparation resources
- Access counselling information
- Compare colleges
- Take mock tests
- Predict admission chances

The platform should be scalable to 100,000+ monthly users.

Tech Stack

Frontend

- Next.js 16 App Router
- TypeScript
- TailwindCSS
- Shadcn UI
- React Query
- Zod

Backend

- Next.js Route Handlers
- Prisma ORM
- PostgreSQL

No separate Express backend.

Single monolithic application.

Infrastructure

Database

PostgreSQL

Provider:

- Neon or
- VPS PostgreSQL

File Storage

Cloudflare R2

Used for:

- PDFs
- Study material
- Mock test assets
- Images

Email

Resend

Used for:

- Notifications
- Newsletter
- Password reset

CDN

Cloudflare

Used for:

- DNS
- Caching
- Image optimization
- Security

User Roles

Guest

Can:

- Search exams
- View colleges
- View papers
- Read resources

Cannot:

- Bookmark
- Subscribe
- Comment

Student

Can:

- Bookmark papers
- Subscribe notifications
- Track exams
- Save colleges

Admin

Full system access

Super Admin

System management Admin management

Core Modules

Authentication Module

Features:

- Register
- Login
- Logout

- Forgot Password
- Email Verification

Database Tables:

User Session VerificationToken

Exam Module

Stores LEET examinations.

Examples:

- IPU LEET
- Haryana LEET
- SLIET SET

Fields:

- id
- name
- slug
- state
- eligibility
- syllabus
- examPattern
- officialWebsite
- applicationStart
- applicationEnd
- examDate

SEO URL:

/exams/ipu-leet

Notification Module

Stores important updates.

Fields:

- title
- description
- examId
- publishDate

Types:

- Form Release
- Admit Card
- Result
- Counselling

SEO URL:

/notifications/...

Previous Year Paper Module

Fields:

- id
- title
- examId
- subject
- year
- pdfUrl
- fileSize
- downloads

Storage:

Cloudflare R2

SEO URL:

/papers/ipu-leet/2020/mathematics

College Module

Fields:

- collegeName
- state
- university
- city
- fees
- placementData
- branches

SEO URL:

/colleges/dtu

Branch Module

Examples:

- CSE
- IT
- ECE
- Mechanical

Linked to colleges.

Counselling Module

Stores:

- counselling schedules
- seat matrices
- cutoff history

SEO URL:

/counselling/ipu

Mock Test Module

Question Types:

- MCQ
- Numerical

Features:

- Timer
- Auto Submit
- Result Analysis

Tables:

MockTest Question Option Attempt Result

Resource Module

Stores:

- Notes
- PDFs
- Preparation guides

Categories:

- Mathematics
 - Physics
 - Engineering Basics
-

Blog Module

Content:

- Exam Guides
- College Reviews
- Counselling Articles

SEO Critical.

Admin Dashboard

Sections:

Dashboard Users Exams Notifications Papers Colleges Resources Blogs Mock Tests

Paper Upload System

Single Upload

Bulk Upload

CSV Import

Folder Upload

Workflow:

Upload PDF → Store in R2 → Save Metadata → Generate Search Index

Search System

Global Search

Search:

- exams
- papers
- colleges
- blogs

Future: Algolia

Initial: PostgreSQL Full Text Search

SEO Architecture

Every page must have:

- metadata
- open graph
- sitemap entry
- structured data

Generate:

/sitemap.xml /robots.txt

Target:

Google search traffic.

Analytics

Track:

- downloads
- searches
- page views

- popular exams
- popular colleges

Admin dashboard charts.

Security

Rate limiting

CSRF protection

Input validation

File validation

Admin RBAC

Audit logs

Database Relationships

User — Bookmarks — Subscriptions — MockAttempts

Exam — Notifications — Papers — Resources — Counselling

College — Branches

MockTest — Questions

Question — Options

Future Features

Rank Predictor

College Predictor

AI Study Assistant

Discussion Forum

Mobile App

Regional Language Support

Push Notifications